EGGS: Expert-Guided Framework Constructed from Semantic Priors for Large-Scale Scene Reconstruction

Anonymous CVPR submission



Figure 1. Comparison of EGGS (ours) with OctreeGS and CityGSv2. EGGS achieves clearer and more faithful detail reconstruction across large scene. (Higher PSNR values reflect higher-fidelity reconstruction)

Abstract

Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS) have demonstrated strong capability in reconstructing 3D scenes ranging from indoor to outdoor environments. For large-scale scenes, reconstruction is often accelerated through parallelization based on geographic partitioning. However, excessive scene size introduces viewpoint locality, leading to two major challenges: reconstruction isolation and redundant computation. The former causes significant inconsistency in rendering quality across viewpoints and the latter leads to repetitive optimization of similar elements, resulting in wasted computational resources. Inspired by the expert optimization principle in Mixture-of-Experts (MoEs), we propose EGGS (Expert-Guided Framework Constructed from Semantic Priors for Large-Scale Scene Reconstruction), an expert-guided framework that replaces geographic partitioning with expert collaboration constructed from semantic priors. EGGS assigns experts to scene elements based on semantic and feature similarity, enabling semantic-aware collaboration that effectively alleviates isolation and redundancy. Extensive experiments on both synthetic and real-world datasets demonstrate that EGGS achieves state-of-the-art reconstruction quality, outperforming existing methods by up to 5% in PSNR. Code and datasets will be released upon acceptance.

1. Introduction

NeRF [1, 25] and 3DGS [9, 13] have demonstrated efficient 3D reconstruction capabilities across diverse scales, ranging from small indoor scenes [22] to large outdoor environments, supporting downstream tasks such as SLAM and AR content creation [37]. Current large-scale reconstruction methods [17, 21, 38, 42, 45] typically adopt geographic region-based parallel algorithms, where each subregion is independently optimized to accelerate reconstruction. However, these methods fail to address the viewpoint locality problem inherent in large-scale scenes. Unlike indoor settings with near-complete scene capture, each viewpoint

in outdoor scenes only observes a partial area, leading to three major issues.

First, **Reconstruction Isolation**. Each subregion reconstruction relies on the viewpoints that observe its internal content. Since viewpoints only provide limited coverage within individual regions, reconstruction quality varies significantly across subregions, an issue unique to large-scale reconstruction. Recent work [20] attempts to improve inter-region consistency via additional loss constraints, yet viewpoint locality remains unresolved: high-quality subregions fail to enhance adjacent low-quality ones, ultimately degrading overall reconstruction fidelity.

Second, **Redundant Computation**. Geographic partitioning separates semantic entities such as buildings, vegetation, and roads; due to viewpoint locality, these elements are repeatedly "rediscovered" and "relearned". This scheme overlooks semantic and structural redundancy, causing repetitive subregion reconstructions that raise computational costs and reduce overall reconstruction quality.

Third, **Multi-Expert Collaboration and Competition**. Modular-GS [34] introduce MoEs [31] into 3DGS for semantic space decomposition. Inspired by the expert division strategy in MOEs for 3D reconstruction, we propose that an expert-guided structure can go beyond semantic segmentation and serve as an alternative to geographic partitioning. However, the multi-experts structure naturally involves expert competition and cooperation, which necessitates designing collaborative mechanisms to prevent boundary ambiguity that could impair rendering quality.

To address these challenges, we present **EGGS**, an Expert-Guided Framework Constructed from Semantic Priors for Large-Scale Scene Reconstruction, which replaces geographic partitioning with a semantic prior multi-expert guided framework. EGGS significantly improves reconstruction balance across large-scale scenes and reducing computational cost for higher-quality reconstruction across the whole scene. At its core, EGGS leverages an expert-guided strategy to balance and accelerate scene reconstruction through three key parts: First, semantic segmentation divides the scene into multiple categories, each assigned to a category expert trained with semantically rich viewpoints for corresponding instance learning and reconstruction. Second, category experts further guide the reconstruction of unselected viewpoints, mitigating reconstruction isolation caused by viewpoint locality and achieving balanced, efficient reconstruction. Third, a collaborative and constraint-based mechanism promotes expert cooperation and suppresses inter-expert conflicts, alleviating boundary-induced blurring.

Extensive experiments in Fig. 6 and Tab. 2 demonstrates that EGGS achieves more balanced and efficient reconstruction than spatial partitioning methods, improving overall reconstruction quality by up to 5% in PSNR. As presented in Fig. 1, EGGS provides more accurate and detailed reconstructions than existing large-scale approaches. Our main contributions are summarized as follows:

1. We propose a novel expert-guided framework for large-scale 3D reconstruction, addressing rendering isolation and redundant computation inherent in geographic partitioning methods. To our best knowledge, this is the first work to apply expert guidance for large-scale reconstruction.
2. We design a complete expert construction and training pipeline that effectively improves reconstruction balance while reducing overall computational cost.
3. We introduce a contribution-weighted expert collaboration algorithm that mitigates boundary ambiguity caused by inter-expert competition.
4. Extensive experiments on both synthetic and real-world datasets validate that EGGS improving overall reconstruction quality by up to 5% in PSNR while improving balance and efficiency.

2. Related Work

2.1. Large-scale Scene Reconstruction

The conflict between high reconstruction costs and low accuracy in large-scale scene reconstruction, driven by scene complexity, remains a key challenge in 3D reconstruction. Recent NeRF variants using heuristic partitioning [38, 39, 44] or voxel grids [2, 6, 26, 45] have supported performance in large-scale scenes but are limited by long training times. Alternatively, 3D Gaussian Splatting (3DGS) has introduced methods like progressive [14, 17, 49] and level-of-detail (LOD) schemes [21, 30] to balance quality and efficiency. However, these geographically partitioned methods still suffer from inconsistent reconstruction quality due to viewpoint limitations, leading to degraded overall results.

2.2. Semantic-awareness

With advancements in 2D semantic understanding [15, 29], recent research has explored semantic integration in 3D reconstruction [3, 8, 28, 47]. Some methods embed semantic information into explicit 3D Gaussian representations by projecting 2D semantic features or aggregating semantic pools [7]. Others leverage external semantic models to guide 3DGS training or inference [51], such as introducing semantic viewpoints consistency losses to mitigate overfitting [11, 41]. These efforts highlight the potential of semantic-aware methods in 3D reconstruction [27], yet their application in complex, large-scale scenes requires further exploration. Inspired by the growing field of 3D spatial semantic awareness, we aim to leverage rich semantic information in large-scale scenes to overcome viewpoint limitations and enable optimization regardless of spatial distance.

2.3. MoEs and Multi-Expert Framework

MoEs framework [10, 12, 35]enables parallel training through expert specialization. Recent successes in visual imaging [31, 46] have inspired its application in 3D reconstruction. Existing work has combined MoEs with 3D reconstruction [4, 5, 19, 34] and made improvements for large-scale scenes [24, 50]. However, prior works have largely ignored the role of experts as reusable semantic priors capable of guiding similar objects that frequently appear in large-scale scenes.

3. EGGS

This section introduces the working mechanism of EGGS. Section 3.1 provides an overview of EGGS. Section 3.2 including the prioritized construction of expert models and their approaches to addressing isolation defects and redundant computation. Section 3.3 describes a fusion mechanism for integrating

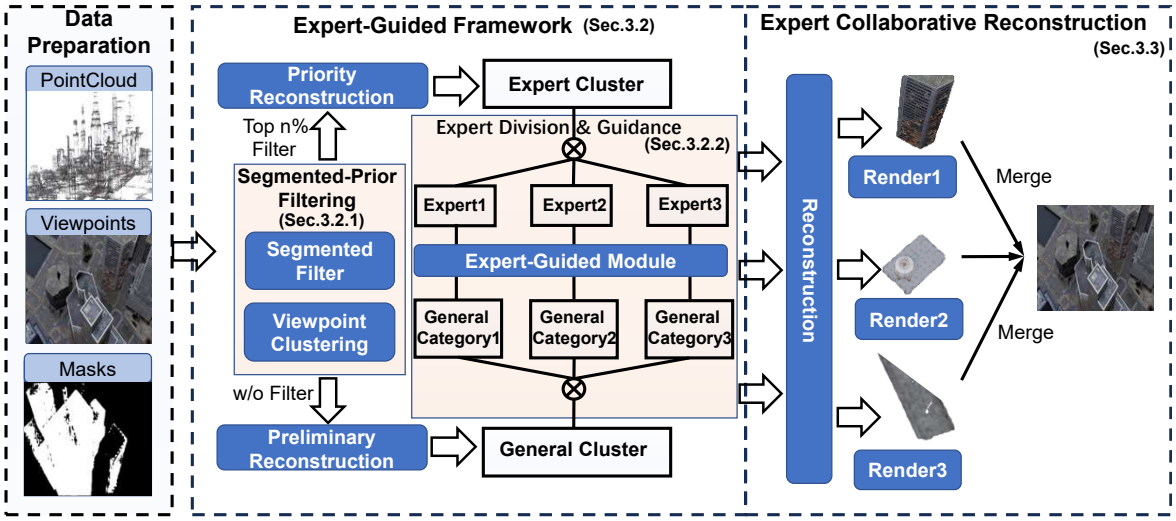


Figure 2. EGGS includes a Semantic-Prior Filtering module for selecting expert models from semantically rich viewpoints and an Expert-Guided module that enhances overall reconstruction via stable matching and GNN.

expert model structures, promoting collaboration and seamless integration among expert models.

3.1. Overview

EGGS is a large-scale collaborative reconstruction framework based on a semantic partitioned multi-expert model. Instead of dividing scenes into geographic blocks, EGGS establishes a complete reconstruction pipeline through two major systems: an expert-guided framework and expert collaborative reconstruction. These correspond to three key modules: Semantic-Prior Filtering, Expert Division & Guidance and Expert Collaborative Reconstruction.

The semantic prior filtering module employs a dual filtering mechanism based on semantics and clustering to select viewpoints and corresponding point clouds that are semantically rich and exhibit strong multi-view consistency from large-scale scenes. These selected viewpoints are prioritized during reconstruction, producing high-quality Gaussian anchors that form category-specific experts representing different semantic classes.

The expert-guided reconstruction module treats these experts as reconstruction priors. During the preliminary reconstruction stage, EGGS performs stable matching between experts and spatially scattered yet semantically consistent elements across the scene, enabling one-to-one expert guidance. Through a graph neural network, prior knowledge is propagated from experts to guide the reconstruction of these elements, achieving high-quality and balanced results. This design effectively mitigates the reconstruction quality disparity and redundant computation "rediscovered" and "relearned") caused by viewpoint locality and isolated rendering of repeated structures.

Finally, the expert collaboration mechanism addresses the intrinsic competition–cooperation problem introduced by replacing geographic partitioning with a multi-expert framework. By assigning gradient weights based on rendering quality, EGGS promotes expert cooperation during reconstruction. Additionally, an opacity-based inter-expert constraint reduces optimization conflicts at expert boundaries, alleviating edge blurring and ensuring coherent global reconstruction.

3.2. Expert-Guided Framework

3.2.1. Semantic-Prior Filtering.

Semantic-prior filtering module of EGGS takes camera viewpoints, point clouds, and semantic segmentation masks as input to generate a subset of viewpoints and point clouds prioritized for expert reconstruction. This module incorporates a Segmented-Prior filtering mechanism consisting of Segmented Filter and viewpoint clustering. The semantic segmentation mask prior is obtained using the CLIP+SAM[15, 29] model guided by textual prompts to capture semantic regions, as detailed in the supplementary material, including model parameters and settings. Semantic score filtering selects the viewpoints and point clouds that best represent expert-specific reconstruction regions, while viewpoint clustering ensures viewpoints consistency within each expert model.

Step 1: Segmented Filter. To quantify the scene information richness of all semantic categories within each viewpoint and prioritize viewpoints that capture more structural details for high-quality expert reconstruction, we define the semantic score as a measure of the semantic richness captured by a single viewpoint. This score is computed through the interaction between the 3D point cloud constructed via SfM[33] and the semantic masks associated with each viewpoint.

Each 3D point has a unique index pid_j and associated 2D keypoint coordinates xys $\in \mathbb{R}^{N_i \times 2}$, where N_i denotes the number of viewpoints in which the point is visible. These coordinates represent the projection positions and visibility of each 3D point across viewpoints. For a given viewpoint cam_i, its semantic score s_{cam_i} is obtained by intersecting all semantic masks of that viewpoint with its corresponding keypoint coordinates and summing over the projected counts:

$$s_{cam_i} = \sum_{k \in K} \sum_{j} pid_j \ \ when \ \ xys_j \in M_k. \tag{1}$$

Here, K denotes the set of all semantic categories, $k \in K$ represents the k-th semantic subcategory, and M_k is the semantic mask of category k in viewpoint cam_i. A higher semantic score s_{cam_i} indicates that the viewpoint contains richer semantic information across all categories, making it more favorable for high-quality reconstruction. Therefore, s_{cam_i} serves as a crucial criterion for selecting viewpoints assigned to expert models.

Step 2: Viewpoint Clustering. Relying solely on semantic richness to select viewpoints and point clouds for expert reconstruction may compromise viewpoints consistency, leading to severe viewpoint locality issues. To address this, we introduce an additional *viewpoint clustering* stage based on the K-Means algorithm, which partitions all viewpoints into multiple clusters according to their camera parameters.

Within each cluster, viewpoints are ranked by their semantic scores s_{cam_i}. A threshold hyperparameter $Top_n\%$ is then applied to select the top $n\%$ viewpoints and their corresponding point clouds for expert-prior reconstruction. This process ensures that each expert receives semantically rich yet spatially diverse inputs, thereby preserving viewpoints consistency and improving reconstruction balance across large-scale scenes.

3.2.2. Expert Division & Guidance.

Expert Construction. The reconstruction isolation and redundant computation issues caused by viewpoint locality essentially stem from the absence of a unified global paradigm to guide and constrain reconstruction across spatially distributed objects. Under limited viewpoint coverage, identical structures cannot be consistently captured from a global perspective, resulting in fragmented reconstructions.

To address this, EGGS constructs multiple category-specific experts using the semantically rich viewpoints and corresponding point clouds obtained from the semantic prior filtering module. These selected data are first reconstructed through a priority reconstruction process, generating high-quality Gaussian anchors that serve as the foundation of each expert. Each category expert is composed of Gaussian anchors corresponding to its semantic category and acts as a global prior to guide consistent reconstruction across the entire scene, as illustrated in Fig. 2.

Stage 1: Multi-Scale Stable Matching. The stable matching algorithm connects the multi-category experts obtained from priority reconstruction with their corresponding semantic instances across the scene, regardless of geographic separation. By establishing associations purely based on feature similarity rather than spatial proximity, this process eliminates reconstruction isolation caused by inconsistent reconstruction of semantically identical objects distributed across large-scale scenes.

Considering that the classical stable matching algorithm has a computational complexity of $O(n^2)$, performing global feature association among all Gaussian anchors is computationally prohibitive. To address this, we propose a multi-scale stable matching scheme based on an Octree hierarchical structure, which reduces the overall time complexity from $O(n^2)$ to $O(n \log n)$, thus meeting the efficiency requirements of expert-guided reconstruction in large-scale scenarios, as shown in Fig. 3.

Specifically, for each semantic class, the algorithm iteratively performs stable matching within Octree sub-blocks, pairing Gaussian anchors from category-specific experts with those of scene

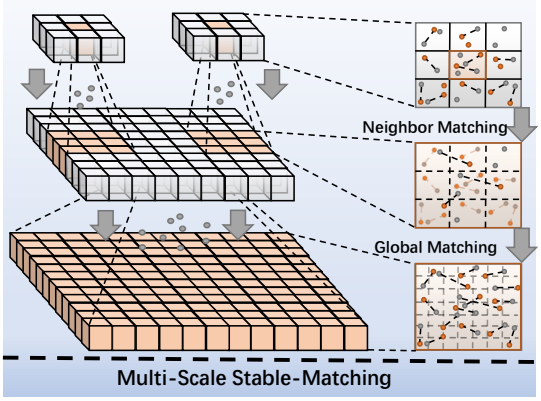


Figure 3. The multi-scale stable matching starts within Octree sub-blocks. Unmatched anchors expand to wider or global scopes. Low-similarity matches are reset and re-evaluated across all experts.

instances exhibiting similar feature representations obtained from preliminary reconstruction, as illustrated in Fig. 2. This matching process can be formulated using a bijective function μ_{unopt}, ensuring that each Gaussian anchor $a \in unopt$ from the scene-object Gaussian anchor set (unopt) is paired with a corresponding Gaussian anchor $b \in opt$ from the expert (opt) of the same semantic category:

$$\mu_{\text{unopt}}(a) = \arg \max_{b \in S_{opt}} s_{unopt}(b), \tag{2}$$

subject to the following constraint:

$$s_{opt}(a) > s_{opt}(\mu_{\text{unopt}}(a)) \cap s_{unopt}(b) > s_{unopt}(\mu^{-1}_{unopt}(a)), \tag{3}$$

where s_{opt} and s_{unopt} denote the similarity preference matrices for expert Gaussian anchors and scene Gaussian anchors of the same category, respectively. The similarity is measured based on the spatial feature embeddings captured by each Gaussian anchor. Since the distributions of opt and $unopt$ anchors within Octree sub-blocks are not always uniform, the hierarchical multi-level matching ensures that any unmatched anchors in higher levels can still be matched in lower levels or during global matching, thereby maintaining completeness and stability of the expert guidance process.

Stage 2: Graph Neural Network Optimization. EGGS equips each targeted expert model with a lightweight graph neural network[32, 36, 40] for parameter learning and guidance. Given a Graph Neural Network with node set $V_{gnn} = V_o \cup V_u$, where V_o is the set of opt and V_u is the set of unopt, for nodes $a \in V_u$ and $b \in V_o$, the edge set E is defined as:

$$E_{gnn} = \{(b, a) \in V_o \times V_u \mid \exists M \subseteq V_o \times V_u \text{ s.t. } (b, a) \in M\}, \tag{4}$$

where $S_{ba} : V_o \times V_u \to \mathbb{R}$ is the similarity preference function between nodes $b \in V_o$ and $a \in V_u$, and M is a stable matching relation. Corresponding to the sparse graph relationships constructed for stable matching, a two-layer GAT network is employed, with each layer consisting of four attention heads. Further information and code will be presented in the supplementary material.

In graph neural networks with node set V_{gnn} and edge set E_{gnn}, the optimization of offsets, scales, and rotation matrices expert parameters can guide global scene gaussian anchors which are initialized throught preliminary reconstruction to approximate object geometry rapidly and adapt further to scene-specific variations. The losses for the three parameters are defined as $L_{offsets}$, $L_{scaling}$, and $L_{rotation}$. Specifically, $L_{offsets}$ is computed using Euclidean distance, $L_{scaling}$ is measured via mean squared error (MSE), and $L_{rotation}$ is based on cosine distance between expert and non-expert representations. The overall loss of the graph neural network during expert model optimization is formulated as:

$$L_g = L_{offsets} + L_{scaling} + L_{rotation}. \tag{5}$$

3.3. Expert Collaborative Reconstruction

To address boundary ambiguity and collaboration challenges in multi-expert models, EGGS introduces two mechanisms that convert Gaussian anchors with ambiguous attribution into collaborative bridges across expert domains.

Gradient Contribution Allocation. All expert models share a total loss function, but in each iteration, the loss is distributed according to each expert's rendering contribution weight to enable coordinated optimization. Suppose there are experts corresponding to K semantic categories, with the total loss function denoted as L. The contribution of expert k in each rendering is α_k, represented by Gaussian anchor weights, satisfying $\sum_{k=1}^{K} \alpha_k = 1$. Each expert's parameters are θ_k, the effective update loss is L_{exp}, and the gradient of the total loss with respect to the parameters is $\nabla_{\theta_k} L$. The gradient assigned to each expert is $\nabla_{\theta_k} L_{\text{exp}}$:

$$\nabla_{\theta_k} L_{\text{exp}} = \alpha_k \nabla_{\theta_k} L. \tag{6}$$

Boundary Gaussian anchors Collaboration. For Gaussian anchors at the boundaries of expert model view frustums, where multiple experts may render and optimize these anchors differently, EGGS adopts a gravitational model from classical mechanics to quantify the influence of each expert's boundary Gaussian anchors on others within the region, facilitating multi-expert collaboration. EGGS uses the Mahalanobis distance $d_{Mah}(i,j)$ to consider the shape and rotation of Gaussian anchors, defining $\mu_j - \mu_i$ as the direction vector:

$$d_{Mah}(i,j) = \sqrt{(\mu_j - \mu_i)^T \sum_{\text{ij}}^{-1} (\mu_j - \mu_i)}, \tag{7}$$

which yields the formula for the influence $F_{i\to j}$ of an expert's boundary Gaussian anchors on co-located Gaussian anchors of other experts:

$$F_{i\to j} = \frac{\alpha_i \cdot \alpha_j}{d_{Mah}(i,j)^2} \cdot (\mu_j - \mu_i), \tag{8}$$

where α_i and α_j represent the opacity of two Gaussian anchors, here extended to denote their mass weights, and x_i and x_j are the center positions of these Gaussian anchors.

4. Experiment

This section presents the experimental evaluation of EGGS. Section 4.1 introduces the datasets, baseline algorithms and experimental setup. Section 4.2 compares EGGS with existing methods in terms of rendering quality across five datasets. Section 4.3 employs radar charts to demonstrate EGGS's improvements in balancing per-view rendering quality and alleviating isolation caused by perspective locality. Section 4.4 explores the impact of key hyperparameters and presents ablation studies to evaluate the contribution of each component. Finally, Section 4.5 discusses potential improvements and outlines directions for future work.

4.1. Experimental Setup

Dataset Configuration. EGGS is evaluated on five large-scale datasets, including one synthetic urban dataset (MatrixCity[16]), two real-world datasets (Hospital and P1 HuiTian), and two complex scenes (Residence and ArtSci) from the open-source UrbanScene3D dataset[18]. MatrixCity is sampled from the SmallCity dataset, whereas Hospital and P1 HuiTian are real-world datasets collected from actual urban environments.

For benchmarking, EGGS is compared against four SOTA large-scale reconstruction methods: *OctreeGS*, *CityGSv2*, *GrendelGS*, and *ScaffoldGS*. All experiments for EGGS, ScaffoldGS, and GrendelGS are conducted on a single Nvidia V100 GPU, with 30,000 training iterations. Due to its two-stage design, *CityGSv2* is trained using 15,000 coarse iterations followed by 15,000 fine-tuning iterations.

For quantitative evaluation, we adopt PSNR, SSIM[43], and LPIPS[48] to comprehensively assess reconstruction performance. Additionally, to highlight EGGS's effectiveness in addressing **viewpoint-local coverage imbalance** and **redundant computation**, we further report the standard deviation (**Std**) of PSNR across viewpoints as a measure of reconstruction consistency, and the total training time (**min**) over 30,000 iterations as an indicator of computational efficiency. For a fair comparison, all models are trained with identical hyperparameter settings, which are detailed in the supplementary material.

4.2. Rendering Quality

EGGS achieves superior rendering performance on both the synthetic MatrixCity and real-world datasets through multi-expert guided optimization. As shown in Tab. 1 and Fig. 4, its overall reconstruction quality matches or surpasses current SOTA methods, with consistently higher PSNR values up to a **5%** gain on the Hospital dataset. On the Residence dataset, EGGS's SSIM and LPIPS are slightly lower than the best results. We attribute this to the dataset's dense residential layout, where a single viewpoint already covers abundant scene content, yielding a natural effect similar to EGGS's experts. This observation highlights the importance of viewpoint coverage in large-scale reconstruction and explains the close performance between EGGS and other SOTA baselines in this case. For the MatrixCity scene, the relatively low performance of OctreeGS may partially stem from conflicts caused by using unified hyperparameter settings across all methods.

4.3. Perspective Balance Experiment

The core idea of the EGGS expert guidance framework is to leverage the Gaussian anchors of high-quality category experts to guide the optimization of corresponding anchors in the same semantic category, thereby improving the overall reconstruction quality of the entire scene. For large-scale scenes, if the rendering quality across all novel viewpoints is more uniform and maintains high

Table 1. Performance comparison across five scenes for PSNR, SSIM, LPIPS with OctreeGS, Scaffold-GS, CityGSv2, Grendel-GS. Best results are in **bold**, and second-best are underlined.

Method	MatrixCity			Hospital Scene			Huitian P1 Scene			Residence			ART-SCI		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Octree-GS[30]	26.39	0.727	0.439	16.28	0.371	0.605	20.46	0.577	0.503	21.59	0.788	0.202	21.57	0.780	0.268
Scaffold-GS[23]	30.53	0.882	0.234	16.21	0.364	0.615	20.38	0.561	0.529	21.19	0.764	0.233	21.14	0.772	0.276
CityGSv2[21]	31.70	0.910	0.202	16.27	0.336	0.768	20.13	0.497	0.710	21.28	0.780	**0.193**	20.35	0.761	0.283
Grendel-GS[49]	30.25	0.885	0.270	15.59	0.316	0.668	19.19	0.460	0.666	21.56	**0.791**	0.212	20.84	0.766	0.293
EGGS (ours)	**31.91**	**0.912**	**0.198**	**17.14**	**0.432**	**0.434**	**20.53**	**0.594**	**0.468**	**21.89**	0.789	0.224	**22.28**	**0.827**	**0.215**

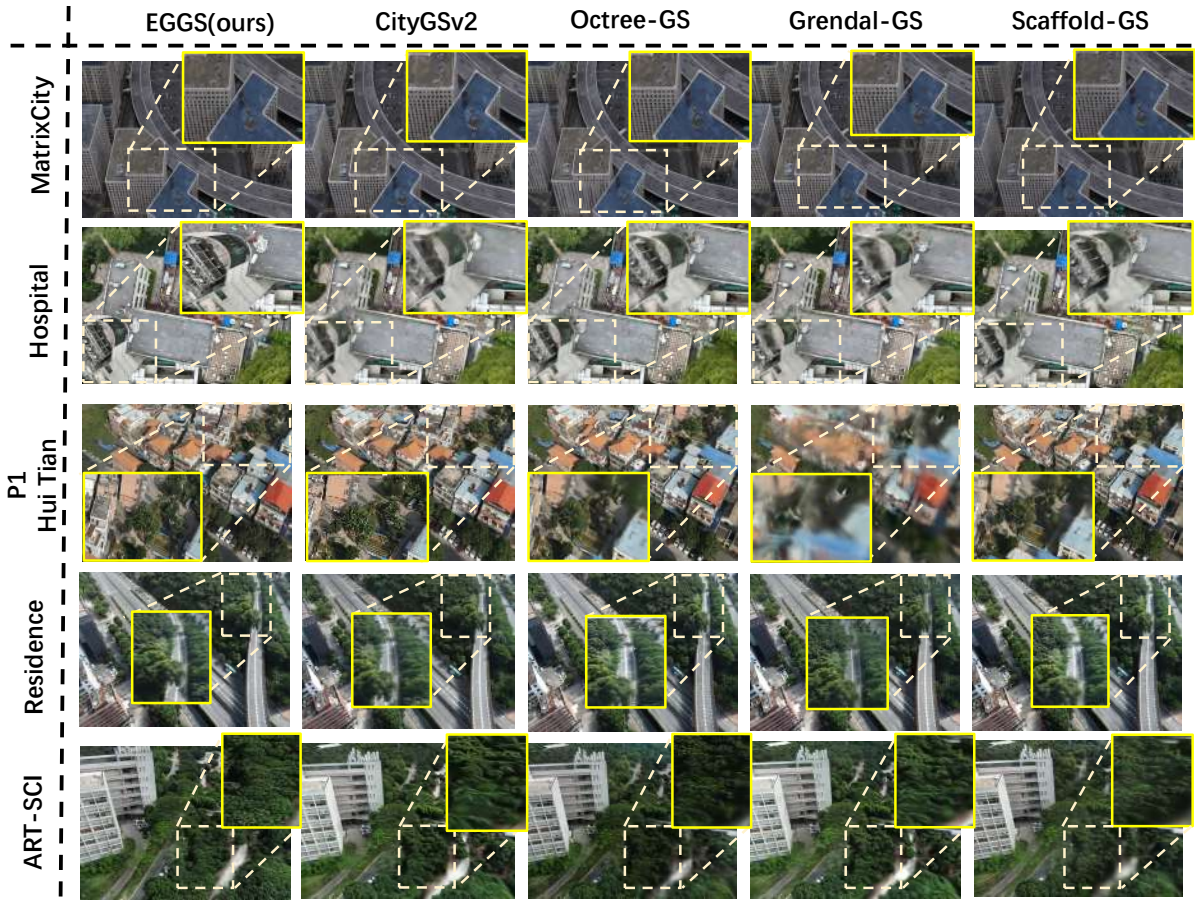


Figure 4. Rendering results of EGGS and baseline methods on MatrixCity and four real-world datasets Hospital, P1 Huitian, Residence and ART-SCI. The magnified regions in views highlight EGGS's superior detail reconstruction, benefiting from expert-guided optimization and scene-wide consistency.

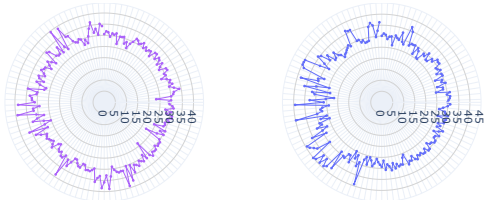
fidelity compared to other methods, it indicates that the expert guidance enhances low-quality views and mitigates reconstruction isolation, resulting in a more balanced and higher-quality reconstruction across the whole scene.

To validate this, we visualize the reconstruction quality distribution of different methods on the MatrixCity dataset using a radar chart, as shown in Fig. 5. A radar plot closer to a circular shape indicates more consistent reconstruction quality across viewpoints. As shown in Fig. 5, CityGSv2 achieves relatively balanced quality on the right-side views but suffers from significant variation on the left. OctreeGS and GrendelGS exhibit even larger deviations, forming irregular, non-circular patterns, implying severe inconsistency between viewpoints. In contrast, EGGS maintains nearly uniform quality across all viewpoints demonstrating that the expert guidance mechanism effectively improves the reconstruction of low quality viewpoints, alleviates viewpoint locality, and enhances the overall reconstruction quality of large-scale scenes.

Beyond qualitative visualization, we further quantify reconstruction consistency across all viewpoints by computing the std of PSNR over five datasets. As shown in Fig. 6 and Tab. 2, methods with high std values generally exhibit stronger reconstruction isolation, which directly manifests as lower or unsta-

Table 2. Performance comparison across five scenes for Std and training time. The standard deviation (std) metric is used to evaluate the consistency of reconstruction quality across viewpoints and training time is used to evaluate the reconstruction efficiency of each algorithm. A smaller or near-one std value indicates uniform rendering quality with minimal reconstruction isolation, while a larger std implies severe viewpoint discrepancy that degrades overall scene reconstruction quality. Best results are in **bold**, and second-best are underlined.

	MatrixCity		Hospital Scene		Huitian P1 Scene		Residence		ART-SCI	
Method	Std ↓	time(min) ↓	Std ↓	time(min) ↓	Std ↓	time(min) ↓	Std ↓	time(min) ↓	Std ↓	time(min) ↓
Octree-GS	4.88	40	1.97	55	0.69	42	1.14	29	0.82	47
Scaffold-GS	2.27	**24**	2.00	46	0.70	**34**	1.28	**17**	1.14	34
CityGSv2	1.02	105	1.66	107	0.56	130	1.55	73	1.65	78
Grendel-GS	3.36	42	2.22	42	1.00	40	1.44	28	1.42	**25**
EGGS (ours)	**0.91**	38	**1.07**	**40**	**0.47**	36	**1.01**	30	**0.47**	32



(a) EGGS Algorithm (b) CityGSv2 Algorithm

(c) Octree-GS Algorithm (d) Grendel-GS Algorithm

Figure 5. Radar charts of PSNR values across different viewpoints for various methods on the same MatrixCity scene. Each spoke represents a view direction, with distance from the center indicating reconstruction quality (higher is better). A more circular shape with smaller variations between points suggests more balanced performance across viewpoints.

ble overall reconstruction quality in Tab. 1. Conversely, methods with lower std tend to deliver more uniform viewpoint performance and consequently more stable overall reconstruction results. *CityGSv2* shows large std in the *Residence*, *ART-SCI*, and *Hospital* scenes—indicating severe isolation—while its lower std on *MatrixCity* aligns with its relatively stronger results in Tab. 1. *OctreeGS* exhibits the highest std in *MatrixCity*, confirming substantial viewpoint inconsistency and matching its degraded reconstruction performance. *GrendelGS* maintains high std across all datasets, consistent with its uniformly suboptimal results and further emphasizing the influence of reconstruction isolation. ScaffoldGS shows a similar pattern, with elevated std that correspondingly limits reconstruction quality. In contrast, EGGS consistently achieves std values close to or below 1 across both synthetic and real-world datasets, reflecting exceptional cross-view uniformity. This indicates that EGGS eliminates reconstruction isolation and delivers stable, high-quality reconstructions across diverse viewpoints and scene regions.

Table 3. Ablation studies on Semantic Segmentation Prior and Semantic-Prior Filtering, Stable Matching, Boundary Collaboration modules.

Method	PSNR ↑	SSIM ↑	LPIPS ↓
EGGS (Full)	31.91	0.912	0.198
w/o Semantic Segmentation	31.89	0.901	0.192
w/o Semantic-Prior Filtering	30.32	0.859	0.347
w/o Stable Matching & GNN	31.28	0.894	0.285
w/o Boundary Collaboration	31.82	0.911	0.210

Additionally, Tab. 2 reports the training time of each method under 30,000 iterations. EGGS requires comparable or even less training time than competing methods, confirming that it not only mitigates redundant computation but also improves reconstruction efficiency in large-scale scenarios.

4.4. Hyperparameter Analysis and Ablation

In EGGS, the hyperparameter top_n controls the expert model's scale. A lower value limits coverage of Gaussian anchors needing guidance, while a higher value causes redundancy and greater overhead. Fig. 7 shows optimal rendering occurs when top_n is between 0.15 and 0.2.

To evaluate the role of each module in the EGGS algorithm, we conducted ablation experiments, and the results are shown in Tab. 3. We first investigate whether the **semantic segmentation prior** has a significant influence on the final reconstruction quality. To this end, we remove the semantic segmentation module and replace it with a strategy that selects experts based on the **information richness** between images and point clouds. As shown in the results, the difference is minor. We attribute this to the fact that semantic segmentation serves as a **suggestive classification mechanism** for category expert selection, while the final expert guidance relies on **feature similarity** and the **multi-scale stable matching algorithm**, which dynamically reassign semantic labels based on high-similarity features. Therefore, possible misclassifications in semantic segmentation have limited impact on the final reconstruction, as the matching process corrects such errors during optimization.

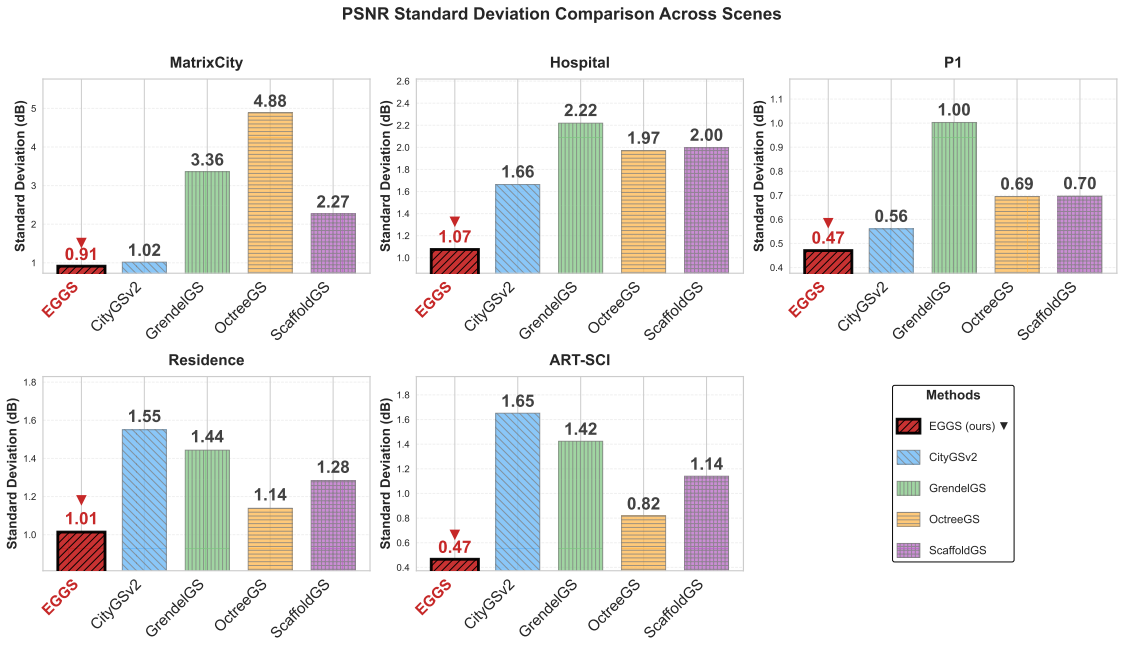


Figure 6. This figure presents the standard deviation of pre-view rendering quality across five large-scale datasets for existing SOTA methods. The deviation values are consistent with the reconstruction quality radar charts and the overall PSNR metrics shown earlier. It can be observed that **EGGS (ours)** exhibits consistently low standard deviation across all scenes, demonstrating its stable and uniform reconstruction quality among different viewpoints.

Next, we remove the semantic prior filtering mechanism and instead randomly sample viewpoints and point clouds for expert construction. This causes a clear performance drop, showing that without semantic priors, the selected viewpoints lack scene information richness and multi-view consistency, resulting in ineffective experts. We then disable the expert guidance mechanism, allowing reconstruction without expert supervision. The decline in quality confirms that expert guidance is essential for stable optimization. Finally, removing the expert collaboration mechanism also degrades rendering quality, demonstrating that insufficient cooperation among experts weakens their collective effectiveness and lowers overall reconstruction performance.

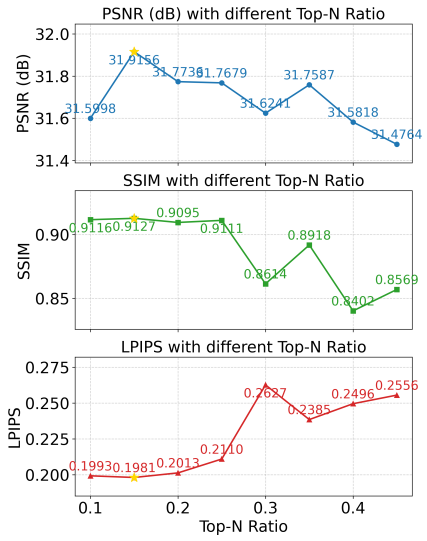


Figure 7. PSNR, SSIM, and LPIPS performance across different Top-N Ratio values.

4.5. Future and Limitation

EGGS demonstrates outstanding performance in addressing the two major challenges posed by the local coverage of large-scale scene perspectives. Currently, the evolving 3D spatial semantic segmentation technology[51] seems to be able to replace the original 2D semantic prior, which had limitations in accuracy, providing EGGS with a more outstanding prior enhancement.

5. Conclusion

This paper presents EGGS, an expert-guided framework for large-scale scene reconstruction. EGGS identifies the viewpoint locality in large-scale scenes, which gives rise to two challenges: reconstruction isolation and redundant computation. By leveraging semantic priors to construct experts and employing an expert-guided reconstruction strategy, EGGS mitigates isolation-induced quality inconsistency and achieves competitive reconstruction efficiency. An expert collaboration mechanism further resolves the boundary ambiguity introduced by experts. Experiments show that existing methods degrade under viewpoint locality, whereas EGGS maintains balanced cross-view reconstruction quality and achieves SOTA performance across diverse datasets, with improvements of up to 5% in PSNR.

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